



Galerkin Reduced Order Models for Compressible Flows with Differentiable Programming

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Accurate and computationally cheap forecasts using Reduced Order Models (ROMs) can facilitate active predictive control strategies to stabilize unsteady flow over airfoils. Galerkin projection based Reduced Order Models (GP-ROMs), derived by projecting the Navier Stokes equations on a truncated Proper Orthogonal Decomposition (POD) basis, have been widely adopted because of their low computational costs and theoretical formulation based on the governing fluid flow equations. However, the traditional GP-ROMs suffer from instabilities and inaccuracies for predictions over long time horizons, in spite of calibration. To address these issues we implement the recently proposed *Neural Galerkin Projection (NeuralGP)* procedure – a differentiable programming-based approach that blends the strengths of the physics driven GP-ROM technique to the purely data driven neural network based techniques previously suggested in literature. The predictive capabilities of the two methods are tested for a transonic flow over a buffeting NACA0012 airfoil. While both techniques give accurate forecasts over short time prediction horizons, only the NeuralGP ROM is able to accurately track the reference POD coefficients over longer prediction horizons with the calibrated GP-ROM diverging in amplitude and frequency. The results highlight the increased accuracy of the NeuralGP technique compared to the calibrated GP-ROM method while being less computationally expensive and more interpretable compared to the traditional physics-agnostic neural networks.

I. Introduction

Incorporating a robust active feedback control for turbulent flows is a promising research problem with the potential to greatly improve the efficiency of flight vehicles, reduce jet noise and design more efficient energy systems [1–4]. The goal of controlling chaotic and nonlinear systems like turbulence, however, requires making predictions several time instants into the horizon in a fast, cheap and accurate manner. High fidelity simulations of fluid flows, while very accurate, can be prohibitively expensive for efficient forecasting and prediction based control techniques. This is particularly true for high Reynolds number turbulent flows, where the number of degrees of freedom can exceed billions for 3D flows and, consequently, can only be simulated on supercomputers.

The increased push towards developing control strategies for turbulent flows has led to several studies focusing on the development of Reduced Order Models (ROM) that retain the accuracy of the high fidelity simulations while greatly reducing the computational resources needed for predictions made over time horizons spanning several non-dimensional flow times [5, 6]. One of the more popular strategies involves the Galerkin projection of the Navier-Stokes equations onto a set of truncated orthogonal basis vectors usually obtained from a Proper Orthogonal Decomposition (POD) of the flow fluctuations. The resulting projected equations can be solved quickly, while capturing the dominant physics of the flow [5, 7–10]. Among these, ROMs for compressible flow problems using the full Navier Stokes equations have not been as widely explored, and several studies have resorted instead to using an isentropic assumption [5] valid only up to moderate Mach numbers. In this study, we follow the work of Bourguet et al. [11] to construct a Galerkin Projection-based ROM (GP-ROM) for the full NS equations, and test it on a transonic flow over an airfoil.

Traditional GP-based ROMs suffer from structural instabilities [5, 12] that severely limit their prediction capabilities and result in model divergence over longer time horizons. This has led to the development of several optimization

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techniques aimed at reducing prediction errors and stabilizing the ROM solutions, like non-linear constrained optimization[13], least squares minimization [14] and Tikhonov regularization [15]. Cordier et al. [15] compared several calibration methodologies and concluded that the technique based on Tikhonov regularization outperformed all other methods. However, despite the application of state-of-the-art calibration techniques, long term predictions from GP-based ROMs remain unreliable [16] necessitating the need for newer methods to improve prediction performance.

In an effort to improve long term predictions from ROMs, recent attempts leverage deep learning-based approaches such as neural networks (NNs) to create data-driven alternatives to GP-ROMs. Several variants of NNs exist, with the LSTM architecture [17] being popular in turbulence [18–21] owing to its strengths in time-series prediction. Although NNs have been reported to have considerable promise, they face important limitations to practical use due to their black-box nature and lack of interpretability. Recently, Mohan et al. [22] developed a hybrid learning approach called *Neural Galerkin Projection (NeuralGP)* that combines Galerkin Projection with deep learning for an isentropic flow, and obtained superior results compared to an isentropic GP-ROM, and without the need for any calibration. The method itself uses *Differentiable Programming* (henceforth referred to as *DiffProg*) - a programming paradigm that enables writing software that can be fully differentiated via automatic differentiation [23–25], and optimized using gradient-descent based methods.

In this study, we seek to implement the Neural Galerkin Projection approach of Mohan et al. [22] for a compressible flow problem of transonic flow over a NACA0012 airfoil and compare its results to the GP-ROM technique of Bourguet et al. [11]. The ultimate goal is to engender a better understanding of the ROMs themselves using a dynamical systems perspective of the ROM-ODE. The rest of the paper is organized as follows: Section II describes the numerical method used for the flow simulation. The details of the grid as well as a brief description of the flow physics is provided in section III. Section IV lays out the POD process along with a description of the leading modes. Section V provides a detailed account of the two methods used to construct the ROMs. Section VI discusses the short and long term predictions from both ROMs. Finally, our contributions are summarized in section VII.

II. Numerical Method

The non-dimensionalized compressible Navier-Stokes equations cast in the strong conservation form are solved in a curvilinear coordinate (ξ, η) frame with J being the Jacobian of the curvilinear transformation given by $\left(J = \frac{\partial(\xi, \eta, \tau)}{\partial(x, y, t)}\right)$ [26].

$$\frac{\partial}{\partial \tau} \left(\frac{\vec{Q}}{J} \right) + \frac{\partial \hat{F}}{\partial \xi} + \frac{\partial \hat{G}}{\partial \eta} = \frac{1}{Re} \left[\frac{\partial \hat{F}_v}{\partial \xi} + \frac{\partial \hat{G}_v}{\partial \eta} \right] \quad (1)$$

where $\vec{Q}$ refers to a vector comprised of the conserved variables. Thus, $\vec{Q} = [\rho, \rho u, \rho v, \rho E]^T$. The vectors $\hat{F}$, and $\hat{G}$ correspond to the inviscid fluxes and the vectors $\hat{F}_v$, $\hat{G}_v$ constitute the viscous fluxes. Thus,

$$F = \begin{bmatrix} \rho \hat{U} \\ \rho u \hat{U} + \xi_x p \\ \rho v \hat{U} + \xi_y p \\ (\rho E + p) \hat{U} \end{bmatrix} \quad \text{and} \quad F_v = \begin{bmatrix} 0 \\ \xi_{x_i} \sigma_{i1} \\ \xi_{x_i} \sigma_{i1} \\ \xi_{x_i} (u_j \sigma_{ij} - \Theta_i) \end{bmatrix} \quad (2)$$

The contravariant velocity component is given by $\hat{U} = \xi_x u + \xi_y v$ and the specific energy density is

$$E = \frac{T}{\gamma(\gamma - 1)M_\infty^2} + \frac{1}{2}(u^2 + v^2) \quad (3)$$

The deviatoric stress (σ_{ij}) and heat flux vector (Θ_i) are given by:

$$\sigma_{ij} = \mu \left(\frac{\partial \xi_k}{\partial x_j} \frac{\partial u_i}{\partial \xi_k} + \frac{\partial \xi_k}{\partial x_i} \frac{\partial u_j}{\partial \xi_k} - \frac{2}{3} \frac{\partial \xi_l}{\partial x_k} \frac{\partial u_k}{\partial \xi_l} \delta_{ij} \right) \quad (4)$$

$$\Theta_i = -\frac{1}{(\gamma - 1)M_\infty^2} \left(\frac{\mu}{Pr} \right) \frac{\partial \xi_j}{\partial x_i} \frac{\partial T}{\partial \xi_j} \quad (5)$$

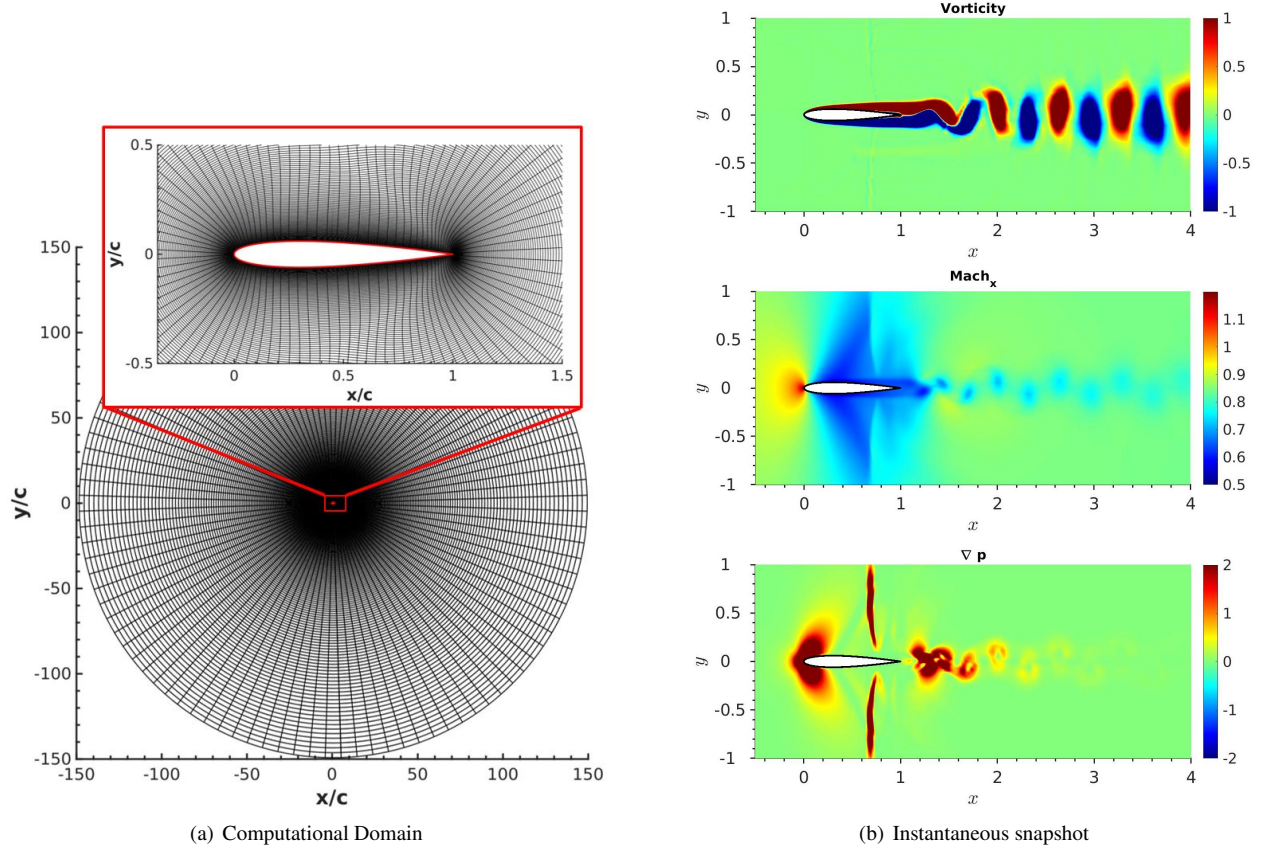


Fig. 1 Details of the computational domain and flow physics shown through an instantaneous snapshot of the flowfield

The $\hat{G}$ and $\hat{G}_v$ vectors have the same structure, with ξ variations replaced by η variations. Stokes' hypothesis is assumed for the bulk viscosity coefficient *i.e* $\lambda = -2/3\mu$. Further details of the derivations may be found in Vinokur [27]. Flow velocities and density are normalized by the free stream velocity (u_∞) and density (ρ_∞), respectively, while the pressure is non-dimensionalized by $\rho_j U_\infty^2$; the perfect gas relationship then reads $p = \rho T / \gamma M_\infty^2$, where M_∞ refers to the freestream Mach number. A constant Prandtl number of 0.72 is assumed along with a constant ratio for the specific heats ($\gamma = 1.4$). Sutherland's law is used to model the temperature dependence of viscosity.

The Roe scheme is used for the inviscid fluxes [28] with a third-order reconstruction using the MUSCL approach [29]. Viscous terms are discretized with a second order central differencing technique. The temporal integration is performed explicitly using a 4th order Runge-Kutta scheme with a time step size of $\Delta t = 0.0001c/u_\infty$. The use of the above methodology along with validation studies for flow past airfoils and cylinders can be found in Ranjan et al. [30]

III. Overall Flowfield Features

A transonic flow past a NACA0012 airfoil at zero angle of attack is chosen for the development of the ROMs. This flowfield has been widely explored using both numerical [11, 31, 32] and experimental [33] techniques in order to analyze the unsteadiness in the wake induced by compressibility effects. At low Mach numbers ($M_\infty < 0.7$), an undulation develops in the wake close to the trailing edge, which transitions into a pair of symmetrical recirculation bubbles with an increase in the freestream Mach number. In the transonic regime ($0.7 < M_\infty < 0.9$), a shock wave develops close to the trailing edge as a result of the acceleration of the flow up to supersonic speeds over the curved airfoil surface. The supersonic region is terminated by an unsteady shock wave, which induces a separation of the boundary layer and leads to the growth of a von Kármán instability. These instability waves result in an alternating vortex shedding resembling a von Kármán vortex street downstream of a cylinder.

Following Bourguet et al. [11], a freestream Mach number of 0.85 and a Reynolds number of $Re = 5,000$ are chosen

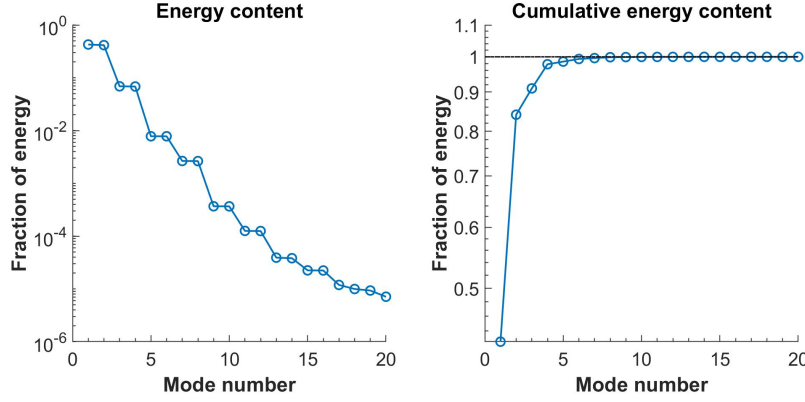


Fig. 2 Fractional as well as the cumulative energy ($\sum_i^{N_{POD}} \lambda_i / \sum_i^{N_t} \lambda_i$) content of the individual POD modes given by eigenvalues of the cross correlation matrix

for the present simulation, which fall within the transonic regime and result in a shock induced unsteadiness in the airfoil wake. The flowfield is simulated using a stretched O-grid comprising of 150,000 points, shown in fig. 1a. The far-field boundaries are at 150 chord lengths in all directions and are modeled using characteristic boundary conditions [34]. Figure 1b shows an instantaneous snapshot of the flowfield. The shock formed towards the trailing edge of the airfoil can be seen clearly in the ∇p contours while the growth of the von Kármán instability resulting in a vortex street in the wake can be observed in the vorticity contours.

IV. Proper Orthogonal Decomposition

The ROM is developed by projecting the full Navier Stokes Equations onto a reduced POD basis. This is facilitated by considering the following transformation of the primitive variables [11]:

$$\{u, v, w, p, \rho\} \longrightarrow q = \{1/\rho, u, v, w, p\} \quad (6)$$

The corresponding spatial POD basis functions ($\Phi_i = \{\Phi_i^{1/\rho}, \Phi_i^u, \Phi_i^v, \Phi_i^w, \Phi_i^p\}^T$) are then obtained using $N_t = 2000$ snapshots of the flowfield sampled $\Delta t = 5 \times 10^{-3}$ following the snapshot POD method of Sirovich [35]. The details of the POD procedure including the spatial and temporal inner products employed are given in appendix A.

Figure 2 shows the eigenvalues of the 20 leading POD modes as a fraction of the energy contained in each mode along with their respective cumulative energy content. A rapid drop in the energy content of the higher POD mode is readily observable; 99.3% of the fluctuation energy is retained with a truncation beyond the 6 leading POD modes. Secondly, the leading modes occur in pairs as is common in flows with dominant convective instabilities. Figure 3(a) and (b) show the leading spatial POD modes for the streamwise velocity and pressure fluctuations; only the odd mode numbers are shown among the repeated mode pairs. The mode shapes represent the convecting vortices in the wake with the primary mode reflecting the spacing of the vortices in the von Kármán vortex street formed in the wake. Their antisymmetric pattern has been observed in previous studies regarding flows with a dominant von Kármán instability [36]. The higher modes represent increasingly finer spatial scales and correspond to superharmonics of the primary vortex shedding frequency.

V. Reduced Order Model

A. Galerkin projection and calibration

A Galerkin projection of the full 3D compressible Navier Stokes equations is performed on the truncated POD basis consisting of the leading $N_{POD} = 6$ modes using the same spatial inner product as used for the POD algorithm (detailed

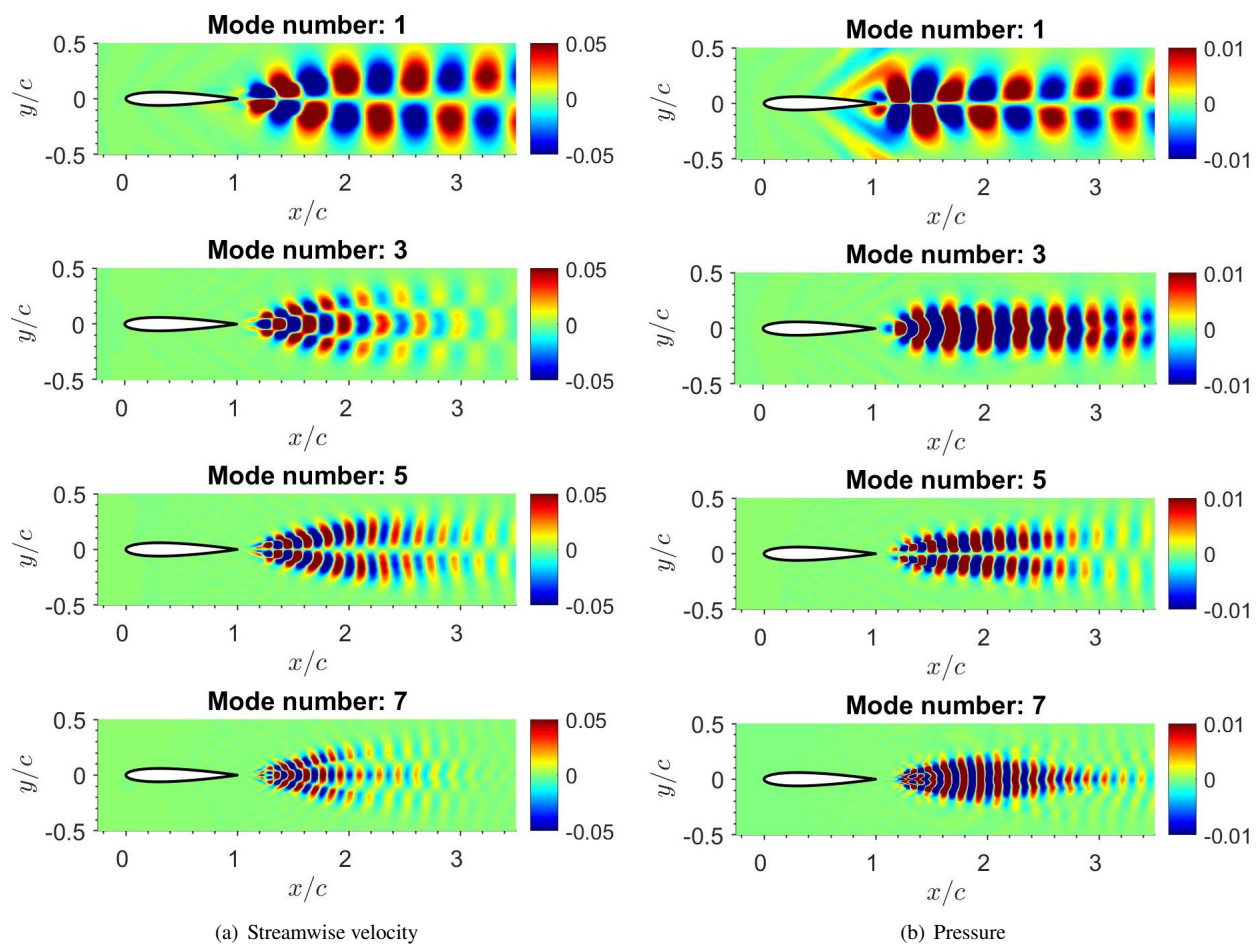


Fig. 3 Leading spatial POD mode shapes for the 7 leading modes

discussion in the context of equation 16 in Appendix IV):

$$\left\langle \frac{\partial \tilde{q}}{\partial t}, \Phi_i \right\rangle_{\Omega} + \left\langle \frac{\partial \tilde{F}_j}{\partial \xi_j}, \Phi_i \right\rangle_{\Omega} = \left\langle \frac{\partial \tilde{F}_j^{vis}}{\partial \xi_j}, \Phi_i \right\rangle_{\Omega} \quad (7)$$

The Einstein summation convention has been used in the above equation. The above Galerkin projection results in the following set of ordinary differential equations for each time coefficient of the leading POD modes:

$$\begin{aligned} \dot{a}_i &= C_i + \sum_{j=1}^{N_{pod}} L_{ij} a_j + \sum_{j=1}^{N_{pod}} \sum_{k=1}^{N_{pod}} Q_{ijk} a_j a_k \\ a_i(0) &= \langle \tilde{q}_0 - \bar{\tilde{q}}, \Phi_i \rangle_{\Omega} \end{aligned} \quad (8)$$

As discussed previously, the time integration of the above model suffers from instabilities [37–40] resulting in a significant prediction error, alteration of oscillation periods, and, finally, a divergence of the temporal integration. In order to obtain a stable prediction that minimizes prediction errors, a calibration method similar to that used by Bourguet et al. [11] is used in the present study. The calibration procedure alters the behavior of the ROM equations using calibration matrices (L_{ij}^{cal} and C_i^{cal}) added to the original ROM coefficient matrices (L_{ij} and C_i) resulting in the following equation:

$$\dot{a}_i = C_i + C_i^{cal} + \sum_{j=1}^{N_{pod}} (L_{ij} + L_{ij}^{cal}) a_j + \sum_{j=1}^{N_{pod}} \sum_{k=1}^{N_{pod}} Q_{ijk} a_j a_k = f_i(L^{cal}, C^{cal}, \mathbf{a}) \quad (9)$$

The calibration matrices are obtained by solving an optimization problem that seeks to minimize the ROM prediction error subject to the constraint of calibration cost. Thus, the following cost function is minimized:

$$\mathcal{J}(C^{cal}, L^{cal}, \theta) = \theta \mathcal{E}(C^{cal}, L^{cal}) + (1 - \theta) C(C^{cal}, L^{cal}) \quad (10)$$

where $\mathcal{E}(C^{cal}, L^{cal})$ and $C(C^{cal}, L^{cal})$ represent normalized ROM error and calibration cost, respectively, blended using the coefficient $\theta \in (0, 1)$.

The choice of error metric and calibration cost has a significant bearing on the calibrated ROM equations and are discussed briefly. The ROM prediction error is given by:

$$E(C^{cal}, L^{cal}) = \sum_{i=1}^{N_{pod}} \int_0^T \left(a_i^{pod}(t) - a_i^{pod}(0) - \int_0^t f_i(C^{cal}, L^{cal}, a^{pod}) dt' \right)^2 dt. \quad (11)$$

The term within the brackets is the difference between the change in the i^{th} time coefficient (a_i) and that predicted by the ROM using the reference POD time coefficients on the RHS of equation 9 over a time horizon $t = 0$ to t . Alternative definitions for the ROM prediction error such as the difference between the instantaneous time rate of changes of a_i s and that predicted by the RHS of the governing equation can also be used [15]. Crucially, the above expression is a linearized definition of the ROM error only taking into account the differences in the rate of change of POD coefficients ($\dot{a}$) versus that predicted by the ROM *at each time step*. While this choice results in a linearization of the optimization problem, it ignores the cumulative effects of the small differences in the predicted $\dot{a}$ over the entire calculated trajectory. The error term E is normalized using the ROM prediction error with no calibration to obtain $\mathcal{E}$ (Detailed discussion in appendix C).

The calibration cost is defined as a measure of the calibration matrices relative to the system dynamic matrices. Here, a sum of an L_2 norm of the C_i vector and a Frobenius norm of the L_{ij} matrix is considered:

$$C(C^{cal}, L^{cal}) = \frac{\|C^{cal}\|^2 + \|L^{cal}\|^2}{\|C\|^2 + \|L\|^2} \quad (12)$$

The calibration procedure then solves the optimization problem to minimize $\mathcal{J}$ for a given value of θ resulting in specific calibration matrices L^{cal} and C^{cal} obtained as a function of the blending parameter θ . A description of the calibration procedure along with its mathematical derivation is given in appendix C.

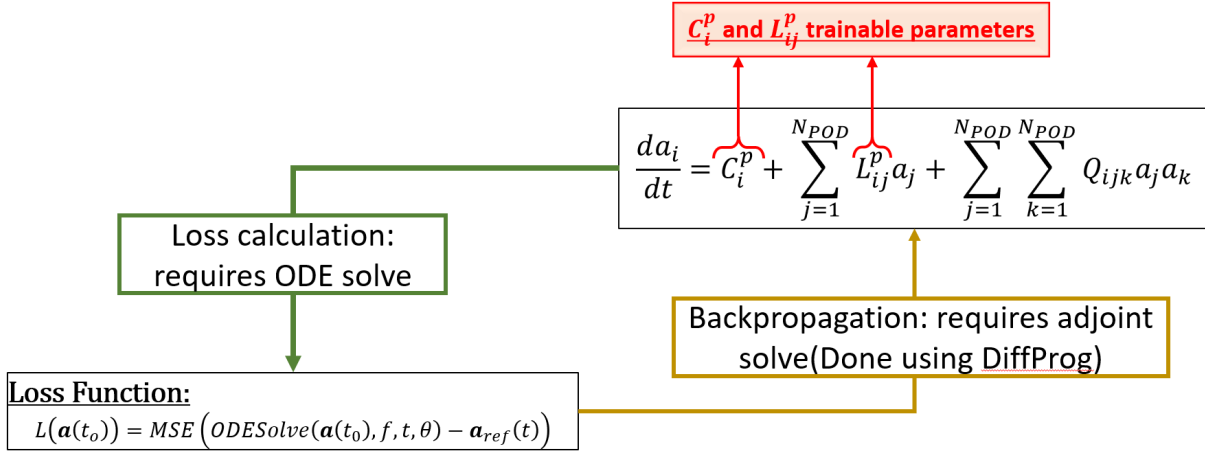


Fig. 4 Schematic of NeuralGP training by treating the linear and constant ROM-ODE coefficients as trainable parameters

B. Differentiable Programming for Neural Galerkin ROM

The GP formulation of equation 8 gives rise to C (Constant), L (Linear) and Q (Quadratic) terms, that are determined via an optimization and calibration process *from the data*. We remark that the data-driven nature of eq. 9, while being structured as an ODE, makes it an excellent candidate to be instead cast as a DiffProg problem. Specifically, we have seen that C and L are the coefficients that have significant impact on the accuracy and long-time stability of the GP-ROM [11, 41], and need considerable manual intervention in the form of calibration. Therefore, our intention is to automate discovery of these coefficients with DiffProg, and directly using the $a_i(t)$ from the high resolution simulation as the training data.

In the DiffProg problem setup, the “unknown” operators in equation 8 are represented as trainable parameters in an NN. The forward pass through the network is defined as a calculation of the $a_i(t)$ trajectories over a stipulated time period (T) using an ODE solver of choice. The network loss is then defined as the L_2 norm of the error between the calculated and real trajectories:

$$L(\mathbf{a}(t_0)) = \sum_{i=1}^{N_{POD}} \int_0^T \left\{ a_i^{ref}(t_1) - \int_0^{t_1} f_i(\mathbf{a}, t) dt \right\}^2 dt_1 \quad (13)$$

$$= \text{MSE}(\text{ODESolve}(\mathbf{a}(t_0), t_0, f, \theta) - \mathbf{a}_{ref}) \quad (14)$$

Where $f_i(\mathbf{a}, t)$ represents the RHS of the ROM-ODE (eq. 8) and a_i^{ref} refers to the reference time coefficients obtained directly from the POD procedure. Crucially, the solution of the ODE problem with given parameters C_i and L_{ij} is an integral part of the loss function and the prediction error is calculated for the entire trajectory over a time period (from $t = 0$ to T). The NN weights, *i.e.* the ROM coefficients, are obtained by back propagating the error through the ODE. The process is outlined in fig. 4. A few key details are listed below:

- Both C_i and L_{ij} are treated as trainable parameters in the NN and they interact in a *physical manner* governed by the form of the projected Navier-Stokes equations.
- DiffProg allows for NN models to be back propagated seamlessly thru the ODE solver via adjoint sensitivity analysis and a combination of forward/reverse mode Automatic Differentiation (AD). The resulting abstract, powerful framework allows for arbitrary neural network models to be tightly integrated within the context of the well-known GP equations.

Thus, contrary to the previously adopted approaches for deep learning-based forecasting tools [18] for fluid flows that use conventional neural networks employing standardized non-linearities, the NeuralGP approach leverages the power of the DiffProg procedure to derive coefficients to a known non-linearity (the ROM-ODE). This results in several key advantages. First, the computational cost of training is extremely low since retention of the Galerkin ODE structure avoids the need of expensive, complex deep NN architectures like LSTMs and CNNs as black boxes for the entire ODE which require several GPUs to train [18, 42, 43]. Secondly, the algorithm attempts to learn a physically

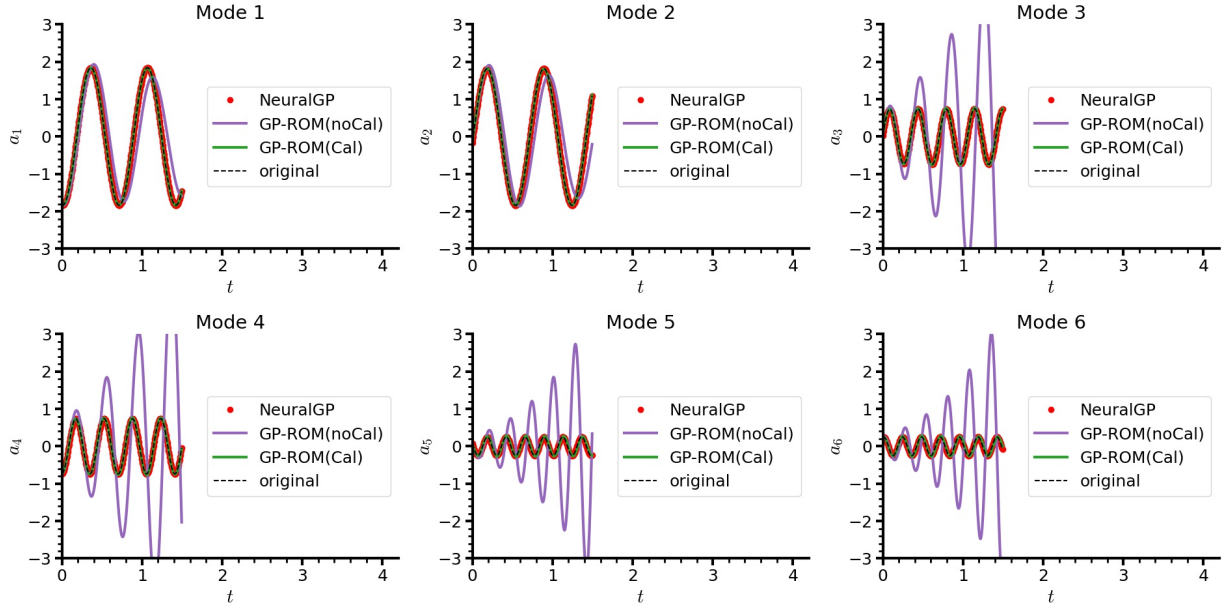


Fig. 5 Time coefficients $a_i(t)$ of the 6 leading POD modes along with predictions from the calibrated and uncalibrated ROMs obtained using Galerkin Projection.

consistent system governing equation rather than fit the data to a generic model for prediction, alleviating the concerns regarding over-training the neural network for a given dataset at the cost of generalizability. This method has been used for an isentropic flow where it has exhibited substantial advantages compared to the previously outlined Galerkin projection/calibration methodology [16].

VI. ROM prediction comparison

The efficacy of the two methods in obtaining stable ROM coefficients are examined using their short and long term predictions. The methods are then later compared based on the properties of the system dynamic matrices obtained from each procedure.

A. Short time prediction

Figure 5 shows the time coefficients ($a_i(t)$) for the six leading POD modes along with predictions from the uncalibrated and calibrated GP-ROMs as well as the NeuralGP-ROM. All of the leading modes time coefficients exhibit a sinusoidal variation with the repeated pairs of eigenvalues discussed in the context of fig. 2 having the same period but with a $\pi/2$ phase lag to maintain orthogonality in time. The frequencies corresponding to the time coefficients also increase with the mode number along with a progressive reduction in amplitude.

A time horizon of $t = 1.5c/u_\infty$ is used for the calibration process which corresponds to two full periods of the leading POD mode. The uncalibrated ROM quickly diverges from the observed POD time coefficients with a rapid growth in amplitudes and an alteration of the oscillation periods most noticeable in the higher POD modes. Even for the two leading POD modes, reasonable results are only obtained over the first oscillation period, after which a significant deviation from the original POD coefficients can be observed. The calibrated ROM results were obtained using a blending parameter of $\theta = 0.9$ (corresponding to a 2.25% calibration cost (eq. 12)). Even with the relatively small additions to the L_{ij} and C_i matrices (as indicated by the small calibration cost), the calibrated ROM tracks the time coefficients obtained from the simulation accurately over the short time horizon. Finally, the results of the NeuralGP ROM also track the original POD time coefficients closely resulting in virtually indistinguishable differences between the calibrated GP and NeuralGP ROM results from the original POD time coefficients.

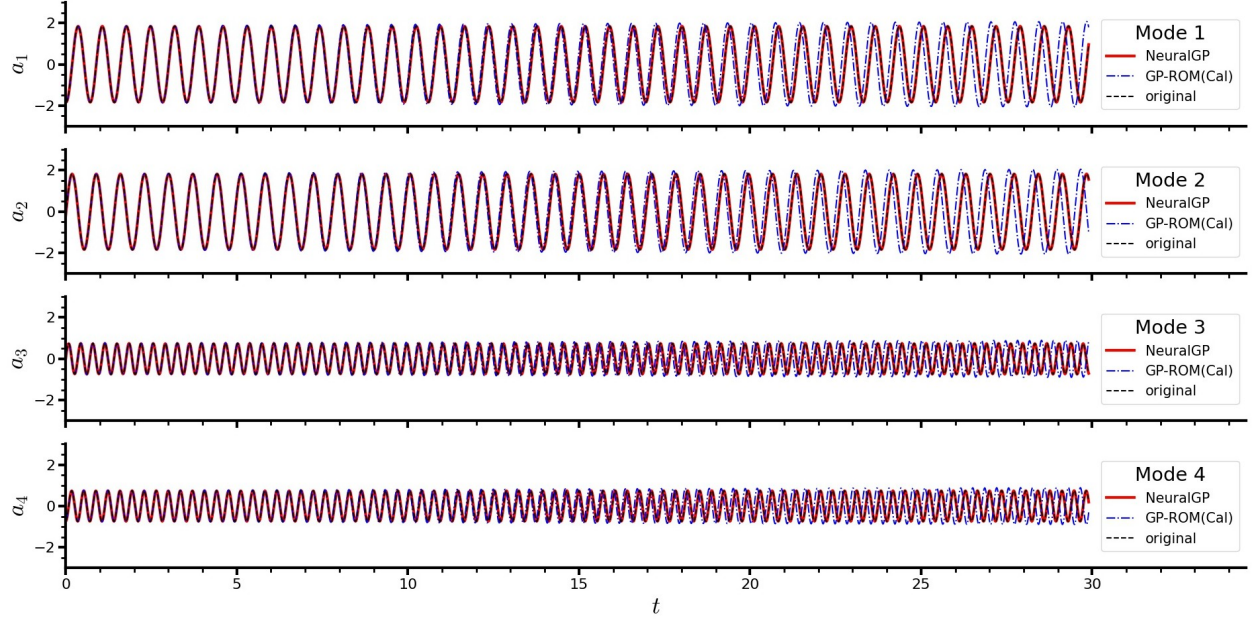


Fig. 6 Long time prediction of the 4 leading POD modes using the calibrated GP-ROM and NeuralGP ROMs

B. Long time predictions

To evaluate the predictive capabilities over a longer time duration, the ROM-ODEs for both the NeuralGP and calibrated ROMs were integrated over a time horizon of $t = 30c/u_\infty$. Figure 6 shows the long time predictions for the four leading modes obtained using both the calibrated and NeuralGP ROMs compared to the original POD time coefficients. The predictions from both ROMs follow the original POD time coefficients closely until $t \approx 10c/u_\infty$. Beyond this time, while the NeuralGP predictions track the original POD time coefficients closely with no indication of divergence, the GP-ROM starts to show significant errors in prediction. The errors manifest chiefly as a change in the oscillation periods along with a slow growth in the amplitude.

The errors in the GP-ROM predictions are further elucidated using Power Spectral Densities (PSD) of the time coefficients within time windows at the beginning and end of the long term prediction horizon. Figure 7 shows the differences in the spectral content of the 3 leading mode pairs within the initial and final time windows. The progressively higher POD mode pairs have correspondingly higher temporal frequencies for which the x-axis of the plots are adjusted. The GP-ROM prediction shows a substantial difference in the spectral content between the two windows tested. Specifically, a shift in the frequency content and amplitude growth is clearly visible in the calibrated GP-ROM predictions in the final time window $t \in (135, 145)$. In contrast, the NeuralGP predictions maintain a near identical spectral distribution irrespective of the time window selected.

VII. Summary and Discussion

ROMs for a compressible, transonic flow over a NACA0012 airfoil are constructed using two different methodologies. First, a more traditional approach using a Galerkin projection of the full Navier Stokes equation on a truncated POD spatial basis is used to obtain a quadratic ODE for the time coefficients of each of the modes. The GP-ROM is found to suffer from instabilities as is commonly reported in the literature. For accurate predictions using the ROM, a calibration is required which is performed here by optimizing linearized ROM prediction errors and calibration cost using a Tikhonov regularization framework. The second method uses machine learning via *differentiable programming* in a manner that seeks to alleviate concerns about the black-box nature of generally used ML based ROMs and improves interpretability and confidence in its predictions. Thus, a hybrid ROM is proposed that blends both ML and Galerkin projection by expressing the ODE usually obtained from the latter as a Differentiable Programming problem. The NeuralGP learns stable coefficients of the Galerkin projection ODE directly from data by treating the ODE coefficients as trainable parameters of a neural network. Compared to traditional black-box neural networks typically used in machine learning-based flow prediction tools (LSTMs or CNNs) the present DiffProg-based approach has several advantages

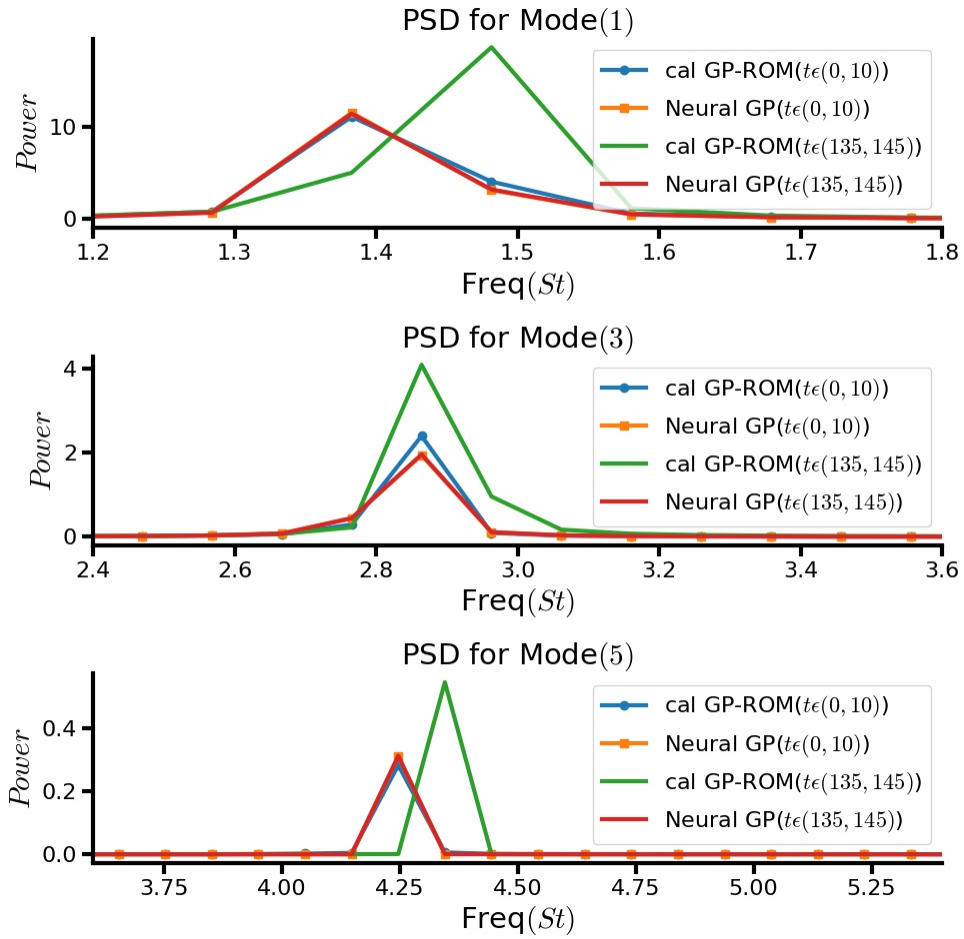


Fig. 7 Comparison of the spectral content of the GP and NeuralGP ROM predictions within initial and final time windows

including lower computational cost, and a mitigation of the risk of over-training for a given data set. Both methods were found to give accurate short time predictions. Over longer prediction horizons, however, the calibrated GP-ROM diverged from the reference POD time coefficients while the NeuralGP continued to give accurate predictions over the longest prediction horizons tested. The data driven nature as well as the greater accuracy in long term integration achieved by the NeuralGP methodology should facilitate the construction of ROMs for more complex flows including fully turbulent flowfields.

Acknowledgements

A.T.M and D.L. are supported by the LDRD (Laboratory Directed Research and Development) program at Los Alamos National Laboratory (LANL) under Contract No. 89233218CNA000001 with the U.S. Department of Energy/National Nuclear Security Administration under project 20190059DR.

A. Details of the POD procedure

The spatial POD basis functions ($\Phi_i = \{\Phi_i^{1/\rho}, \Phi_i^u, \Phi_i^v, \Phi_i^w, \Phi_i^p\}^T$) are sought that maximize the projection of the fluctuation fields ($q(\mathbf{x}, t)$) and optimally represent the second order statistics of the flow:

$$\max_{\Phi(\mathbf{x})} \frac{\overline{\langle q(\mathbf{x}, t), \Phi(\mathbf{x}) \rangle_\Omega}}{\overline{\langle \Phi(\mathbf{x}), \Phi(\mathbf{x}) \rangle_\Omega}} \quad (15)$$

Here, the $\bar{\cdot}$ operator refers to a time averaging operation and the $\langle \cdot \rangle_\Omega$ refers to a carefully chosen spatial inner product defined on the spatial domain Ω , which, for two separate realizations of the state vector (q^I and q^{II}), is given by:

$$\langle q^I, q^{II} \rangle_\Omega = \sum_{ivar=1}^{nvar} \frac{1}{\sigma_{ivar}^2} \int_{\Omega} q_{ivar}^I q_{ivar}^{II} d\mathbf{x} \quad (16)$$

Thus, the contribution of the i^{th} variable of the state vector (q_{ivar}^I) to the spatial inner product is normalized by σ_{ivar}^2 , the spatio-temporally averaged energy of the variable:

$$\sigma_{ivar}^2 = \int_{\Omega} \overline{q_{ivar}^2} d\mathbf{x} \quad (17)$$

Using the above definition of the spatial inner product, the time cross correlation matrix is defined as:

$$K(t_1, t_2) = \langle q(\mathbf{x}, t_1), q(\mathbf{x}, t_2) \rangle_\Omega \quad (18)$$

Following the snapshot POD method [35], the maximization of the flowfield projections (equation 15) is equivalent to solving the following eigenvalue problem to obtain the time variation of each spatial mode:

$$\overline{K(t_1, t) a_i(t)} = \lambda_i a_i(t_1) \quad (19)$$

The above time coefficients ($a_i(t)$) are used to obtain the spatial basis functions by calculating the projection of the simulation snapshots on these time coefficients:

$$\Phi_i(\mathbf{x}) = \frac{1}{\sqrt{\lambda_i}} \overline{\tilde{q}(\mathbf{x}, t) a_i(t)} \quad (20)$$

Finally, the simulation snapshots (q) can be expressed using the time coefficients ($a_i(t)$) and the spatial basis functions ($\Phi_i(\mathbf{x})$) after a decomposition into mean ($\bar{q}$) and fluctuating component ($\tilde{q}$). The optimality of the POD spatial basis then allows for an error bound truncation using only the leading N_{POD} modes:

$$q = \bar{q} + \tilde{q} = \bar{q} + \sum_{i=0}^{\infty} a_i(t) \Phi_i(\mathbf{x}) \approx \bar{q} + \sum_{i=0}^{N_{POD}} a_i(t) \Phi_i(\mathbf{x}) \quad (21)$$

B. Details of the GP-ROM coefficients

The coefficients of the quadratic ROM-ODE (eq. 8) are given by:

$$\begin{aligned} C_i &= \langle F_{11}^\alpha - A_{11}^\alpha, \Phi_i \rangle_\Omega \\ L_{ij} &= \langle F_{1(j+1)}^\alpha + F_{(j+1)1}^\alpha - A_{1(j+1)}^\alpha - A_{(j+1)1}^\alpha, \Phi_i \rangle_\Omega \\ Q_{ijk} &= \langle F_{(j+1)(k+1)}^\alpha - A_{(j+1)(k+1)}^\alpha, \Phi_i \rangle_\Omega \end{aligned} \quad (22)$$

where the Greek indices (α, β) indicate a summation over the coordinate directions. The F_{ij} and A_{ij} matrices are given by:

$$A_{jk}^i = \begin{bmatrix} \Phi_j^{\star u_i} \Phi_{k,i}^{\star(1/\rho)} - \Phi_j^{\star(1/\rho)} \Phi_{k,i}^{\star u_i} \\ \Phi_j^{\star u_i} \Phi_{k,i}^{\star u_1} + \Phi_j^{\star(1/\rho)} \Phi_{k,i}^{\star p} \delta_{1i} \\ \Phi_j^{\star u_i} \Phi_{k,i}^{\star u_2} + \Phi_j^{\star(1/\rho)} \Phi_{k,i}^{\star p} \delta_{2i} \\ \Phi_j^{\star u_i} \Phi_{k,i}^{\star u_3} + \Phi_j^{\star(1/\rho)} \Phi_{k,i}^{\star p} \delta_{3i} \\ \gamma \Phi_j^{\star p} \Phi_{k,i}^{\star u_i} + \Phi_j^{\star u_i} \Phi_{k,i}^{\star p} \end{bmatrix} \text{ and } F_{jk}^i = \begin{bmatrix} 0 \\ \Phi_j^{\star(1/\rho)} \tau_{1ik,i}^\star \\ \Phi_j^{\star(1/\rho)} \tau_{2ik,i}^\star \\ \Phi_j^{\star(1/\rho)} \tau_{3ik,i}^\star \\ \frac{\gamma\mu}{P_r} \left(\Phi_j^{\star p} \Phi_k^{\star(1/\rho)} \right)_{,ii} + (\gamma - 1) \Phi_{j,i}^{\star u_\alpha} \tau_{\alpha ik}^\star \end{bmatrix}. \quad (23)$$

Here, $\tau_{ijk}^\star = \mu(\Phi_{k,j}^{\star u_i} + \Phi_{k,i}^{\star u_j} - 2/3 \Phi_{k,\alpha}^{\star u_\alpha} \delta_{ij})$ and $\Phi^\star$ refers to the augmented POD basis comprising of the mean flow and the spatial POD modes: $\Phi^\star = [\bar{q}, \Phi_1, \dots, \Phi_{N_{pod}}]$. The matrices C_i , L_{ij} , and Q_{ijk} of equation 8 can be pre-computed using only the spatial POD modes. The integration of the ROM equation itself requires $N_{POD}(1 + N_{POD} + N_{POD}^2)$ scalar products per time step and is much cheaper than integrating the governing equations (equation 1).

C. Details of the GP-ROM calibration

For the purposes of calibrating the ROM, a linearized definition for the ROM prediction error is adopted:

$$E(C^{cal}, L^{cal}) = \sum_{i=1}^{N_{pod}} \int_0^T \left(a_i^{pod}(t) - a_i^{pod}(0) - \int_0^t f_i(C^{cal}, L^{cal}, a^{pod}) dt' \right)^2 dt. \quad (24)$$

The error for a given set of calibration matrices (L_{ij}^{cal} and C_i^{cal}) is then normalized using the ROM prediction error without any calibration, *i.e.*, L_{ij}^{cal} and $C_i^{cal} = 0$:

$$\mathcal{E}(C^{cal}, L^{cal}) = \frac{E(C^{cal}, L^{cal})}{E(0, 0)} \quad (25)$$

The calibration cost is defined as a measure of the calibration matrices relative to the system dynamic matrices. Here a sum of an L_2 norm of the C_i vector and a Frobenius norm of the L_{ij} matrix is considered:

$$C(C^{cal}, L^{cal}) = \frac{\|C^{cal}\|^2 + \|L^{cal}\|^2}{\|C\|^2 + \|L\|^2} \quad (26)$$

The calibration coefficients are then obtained by minimizing the following cost function that blends the calibration error with the calibration cost using the blending coefficient θ . This is similar to the regularization parameter in a Tikhonov regularization framework [15, 44].

Minimizing $\mathcal{J}$ is equivalent to solving the following N_{POD} linear systems of size $N_{POD} + 1$:

$$A^{cal} (K_i^{cal})^T = b^i \quad (27)$$

Where $K^{cal} = [C^{cal}, L^{cal}]$ and the matrices A^{cal} and b^i are given as:

$$A_{ij} = \int_0^T \left(\int_0^t a_i^\star dt' \right) \left(\int_0^t a_j^\star dt' \right) dt + \theta^\star \delta_{ij} \quad (28)$$

$$b_i^k = \int_0^T \left(a_k^{pod}(t) - a_k^{pod}(0) - \int_0^t f_k(O, O, a^{pod}) dt' \right) \left(\int_0^t a_i^\star dt' \right) dt, \quad (29)$$

Here, $a^* = [1, a_1, a_2, \dots, a_{NPOD}]$ and

$$\theta^* = \frac{1 - \theta}{\theta} \frac{E(0, 0)}{\|C\|^2 + \|L\|^2} \quad (30)$$

References

- [1] Gad-el Hak, M., “Flow control: The future,” *Journal of aircraft*, Vol. 38, No. 3, 2001, pp. 402–418.
- [2] Bons, J. P., Hansen, L. C., Clark, J. P., Koch, P. J., and Sondergaard, R., “Designing low-pressure turbine blades with integrated flow control,” *Turbo Expo: Power for Land, Sea, and Air*, Vol. 47268, 2005, pp. 1079–1091.
- [3] Samimy, M., Kim, J. H., Kastner, J., Adamovich, I., and Utkin, Y., “Active control of high-speed and high-Reynolds-number jets using plasma actuators,” *Journal of Fluid Mechanics*, Vol. 578, 2007, p. 305.
- [4] Wei, M., and Freund, J. B., “A noise-controlled free shear flow,” *Journal of Fluid Mechanics*, Vol. 546, 2006, p. 123.
- [5] Rowley, C. W., Colonius, T., and Murray, R. M., “Model reduction for compressible flows using POD and Galerkin projection,” *Physica D: Nonlinear Phenomena*, Vol. 189, No. 1-2, 2004, pp. 115–129.
- [6] Noack, B. R., Morzynski, M., and Tadmor, G., *Reduced-order modelling for flow control*, Vol. 528, Springer Science & Business Media, 2011.
- [7] Rempfer, D., “On low-dimensional Galerkin models for fluid flow,” *Theoretical and Computational Fluid Dynamics*, Vol. 14, No. 2, 2000, pp. 75–88.
- [8] Lorenzi, S., Cammi, A., Luzzi, L., and Rozza, G., “POD-Galerkin method for finite volume approximation of Navier–Stokes and RANS equations,” *Computer Methods in Applied Mechanics and Engineering*, Vol. 311, 2016, pp. 151–179.
- [9] Benner, P., Gugercin, S., and Willcox, K., “A survey of projection-based model reduction methods for parametric dynamical systems,” *SIAM review*, Vol. 57, No. 4, 2015, pp. 483–531.
- [10] Chinesta, F., Huerta, A., Rozza, G., and Willcox, K., “Model order reduction,” *Encyclopedia of Computational Mechanics*, 2Eds., Wiley, 2016.
- [11] Bourguet, R., Braza, M., and Dervieux, A., “Reduced-order modeling of transonic flows around an airfoil submitted to small deformations,” *Journal of Computational Physics*, Vol. 230, No. 1, 2011, pp. 159–184.
- [12] Callahan, J. L., Brunton, S. L., and Loiseau, J.-C., “On the role of nonlinear correlations in reduced-order modeling,” *arXiv preprint arXiv:2106.02409*, 2021.
- [13] Bergmann, M., “Optimisation aérodynamique par réduction de modèle POD et contrôle optimal: application au sillage laminaire d’un cylindre circulaire,” Ph.D. thesis, Institut National Polytechnique de Lorraine, 2004.
- [14] Couplet, M., Basdevant, C., and Sagaut, P., “Calibrated reduced-order POD-Galerkin system for fluid flow modelling,” *Journal of Computational Physics*, Vol. 207, No. 1, 2005, pp. 192–220.
- [15] Cordier, L., El Majd, B. A., and Favier, J., “Calibration of POD reduced-order models using Tikhonov regularization,” *International journal for numerical methods in fluids*, Vol. 63, No. 2, 2010, pp. 269–296.
- [16] Mohan, A. T., Nagarajan, K., and Livescu, D., “Learning Stable Galerkin Models of Turbulence with Differentiable Programming,” *arXiv preprint arXiv:2107.07559*, 2021.
- [17] Hochreiter, S., and Schmidhuber, J., “Long short-term memory,” *Neural computation*, Vol. 9, No. 8, 1997, pp. 1735–1780.
- [18] Mohan, A. T., and Gaitonde, D. V., “A deep learning based approach to reduced order modeling for turbulent flow control using LSTM neural networks,” *arXiv preprint arXiv:1804.09269*, 2018.
- [19] Ahmed, S. E., San, O., Rasheed, A., and Iliescu, T., “A long short-term memory embedding for hybrid uplifted reduced order models,” *Physica D: Nonlinear Phenomena*, Vol. 409, 2020, p. 132471.
- [20] Deng, Z., Chen, Y., Liu, Y., and Kim, K. C., “Time-resolved turbulent velocity field reconstruction using a long short-term memory (LSTM)-based artificial intelligence framework,” *Physics of Fluids*, Vol. 31, No. 7, 2019, p. 075108.
- [21] Maulik, R., Lusch, B., and Balaprakash, P., “Non-autoregressive time-series methods for stable parametric reduced-order models,” *Physics of Fluids*, Vol. 32, No. 8, 2020, p. 087115.

- [22] Mohan, A., Nagarajan, K., and Livescu, D., “Learning Physics-based Galerkin models of turbulence with Neural Differential Equations,” *Bulletin of the American Physical Society*, 2020.
- [23] Griewank, A., et al., “On automatic differentiation,” *Mathematical Programming: recent developments and applications*, Vol. 6, No. 6, 1989, pp. 83–107.
- [24] Bartholomew-Biggs, M., Brown, S., Christianson, B., and Dixon, L., “Automatic differentiation of algorithms,” *Journal of Computational and Applied Mathematics*, Vol. 124, No. 1-2, 2000, pp. 171–190.
- [25] Baydin, A. G., Pearlmutter, B. A., Radul, A. A., and Siskind, J. M., “Automatic differentiation in machine learning: a survey,” *Journal of machine learning research*, Vol. 18, 2018.
- [26] Anderson, D., Tannehill, J. C., and Pletcher, R. H., *Computational fluid mechanics and heat transfer*, CRC Press, 2016.
- [27] Vinokur, M., “An analysis of finite-difference and finite-volume formulations of conservation laws,” *Journal of computational physics*, Vol. 81, No. 1, 1989, pp. 1–52.
- [28] Roe, P. L., “Approximate Riemann solvers, parameter vectors, and difference schemes,” *Journal of Computational Physics*, Vol. 43, No. 2, 1981, pp. 357–372.
- [29] Morrison, J. H., “Flux difference split scheme for turbulent transport equations. 1990,” *AIAA Paper*, 1990, pp. 90–5251.
- [30] Ranjan, R., Unnikrishnan, S., and Gaitonde, D., “A robust approach for stability analysis of complex flows using high-order Navier-Stokes solvers,” *Journal of Computational Physics*, Vol. 403, 2020, p. 109076.
- [31] Bouhadji, A., and Braza, M., “Organised modes and shock–vortex interaction in unsteady viscous transonic flows around an aerofoil: Part I: Mach number effect,” *Computers & fluids*, Vol. 32, No. 9, 2003, pp. 1233–1260.
- [32] Bouhadji, A., and Braza, M., “Organised modes and shock–vortex interaction in unsteady viscous transonic flows around an aerofoil: Part II: Reynolds number effect,” *Computers & fluids*, Vol. 32, No. 9, 2003, pp. 1261–1281.
- [33] Harris, C. D., “Two-dimensional aerodynamic characteristics of the NACA 0012 airfoil in the Langley 8 foot transonic pressure tunnel,” 1981.
- [34] Whitfield, D., “Three-dimensional unsteady Euler equations solution using flux vector splitting,” *17th Fluid Dynamics, Plasma Dynamics, and Lasers Conference*, 1984, p. 1552.
- [35] Sirovich, L., “Turbulence and the dynamics of coherent structures. I. Coherent structures,” *Quarterly of applied mathematics*, Vol. 45, No. 3, 1987, pp. 561–571.
- [36] Bergmann, L. C., and Brancher, J., “Optimal rotary control of the cylinder wake using POD reduced order model. AIAA Paper 2004-2323,” *2nd AIAA Flow Control Conference*, 2004.
- [37] Iollo, A., Lanteri, S., and Désidéri, J.-A., “Stability properties of POD–Galerkin approximations for the compressible Navier–Stokes equations,” *Theoretical and Computational Fluid Dynamics*, Vol. 13, No. 6, 2000, pp. 377–396.
- [38] Balajewicz, M., and Dowell, E. H., “Stabilization of projection-based reduced order models of the Navier–Stokes,” *Nonlinear Dynamics*, Vol. 70, No. 2, 2012, pp. 1619–1632.
- [39] Amsallem, D., and Farhat, C., “Stabilization of projection-based reduced-order models,” *International Journal for Numerical Methods in Engineering*, Vol. 91, No. 4, 2012, pp. 358–377.
- [40] Akhtar, I., Nayfeh, A. H., and Ribbens, C. J., “On the stability and extension of reduced-order Galerkin models in incompressible flows,” *Theoretical and Computational Fluid Dynamics*, Vol. 23, No. 3, 2009, pp. 213–237.
- [41] Nagarajan, K. K., Cordier, L., and Airiau, C., “Development and application of a reduced order model for the control of self-sustained instabilities in cavity flows,” *Communications in Computational Physics*, Vol. 14, 2013, pp. pp–186.
- [42] Mohan, A., Daniel, D., Chertkov, M., and Livescu, D., “Compressed convolutional LSTM: An efficient deep learning framework to model high fidelity 3D turbulence,” *arXiv preprint arXiv:1903.00033*, 2019.
- [43] Mohan, A. T., Tretiak, D., Chertkov, M., and Livescu, D., “Spatio-temporal deep learning models of 3D turbulence with physics informed diagnostics,” *Journal of Turbulence*, Vol. 21, No. 9-10, 2020, pp. 484–524.
- [44] Bergmann, M., Bruneau, C.-H., and Iollo, A., “Enablers for robust POD models,” *Journal of Computational Physics*, Vol. 228, No. 2, 2009, pp. 516–538.